SPX Regime Detection Report

Hidden Markov Model — Regime Analysis & Strategy Backtest

Assets: S&P 500 • Nasdaq-100 (QQQ) • Russell 2000 (IWM)

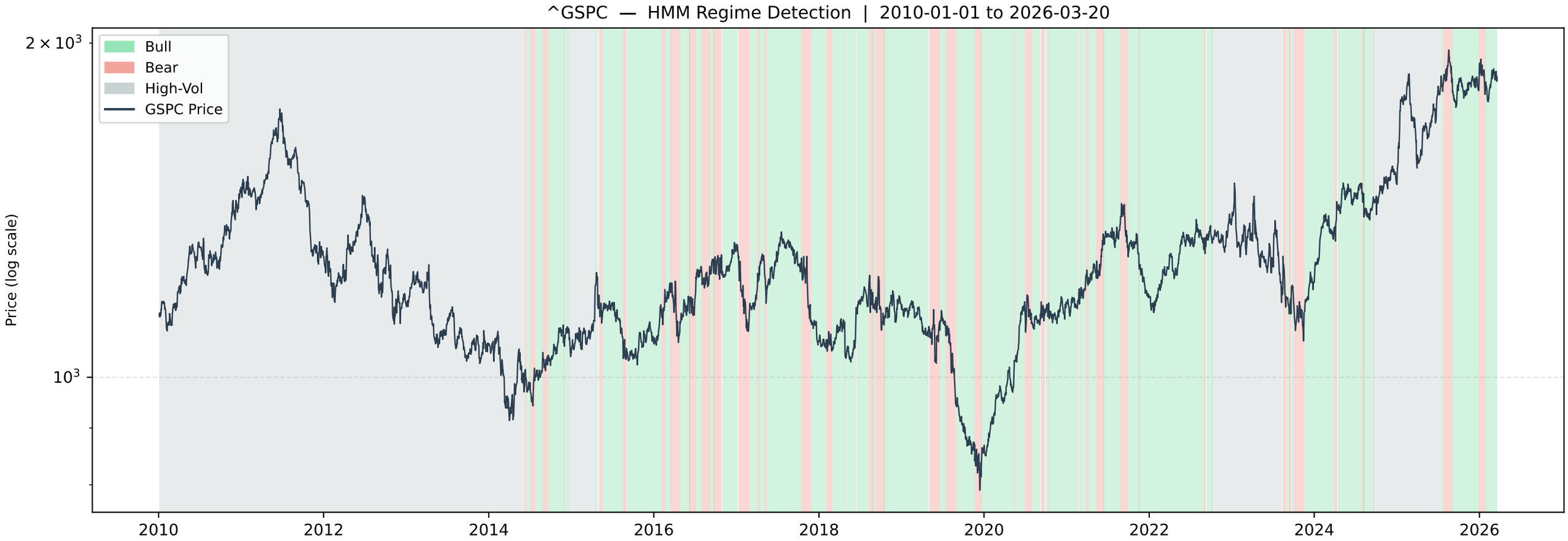
Period: 2010-01-01 → March 20, 2026

Methodology: 3-state Gaussian HMM | Baum-Welch EM | Viterbi decoding
Walk-forward validation | Persistence filter | Multi-strategy backtest

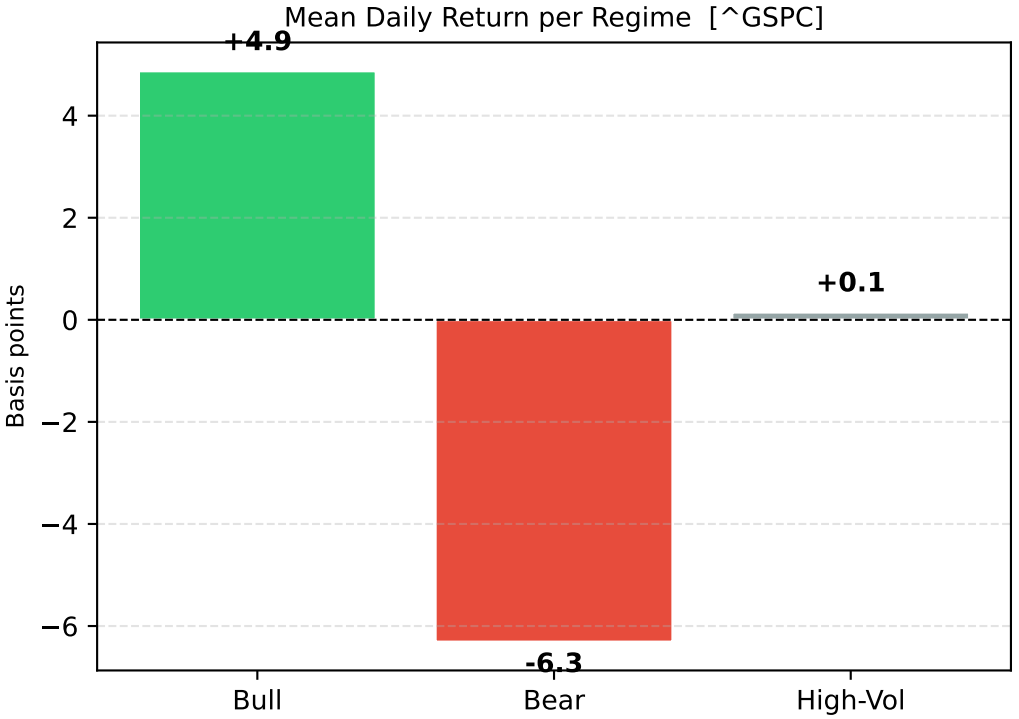
Generated: March 20, 2026

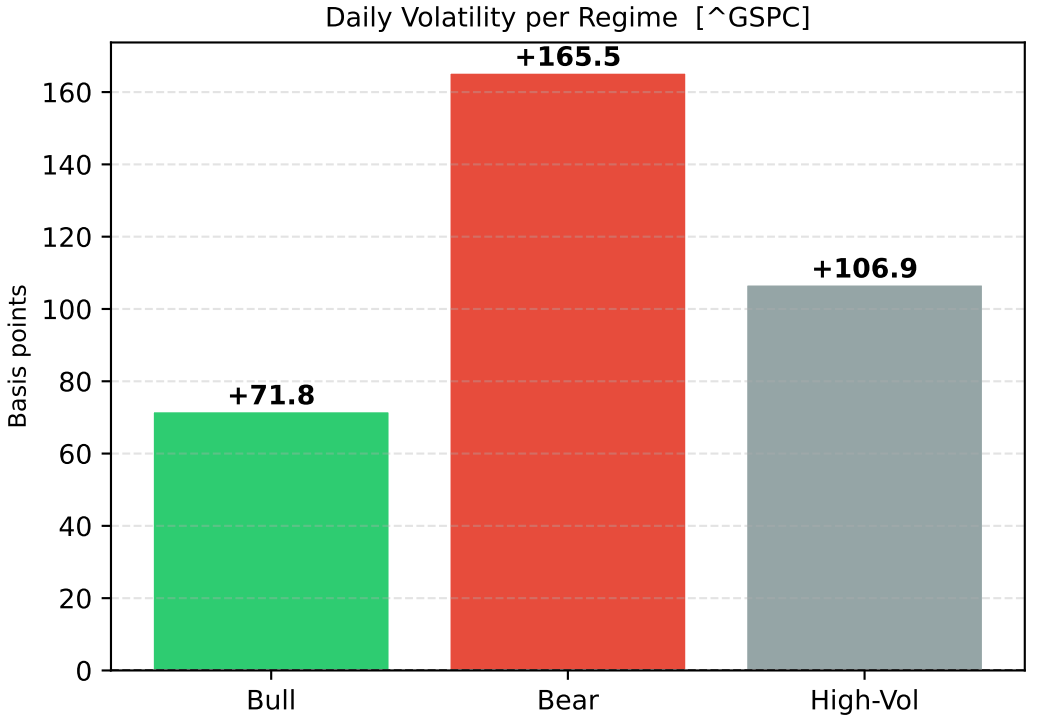
Asset: ^GSPC

2010-01-01 to 2026-03-20 | Calibrated synthetic [^GSPC]

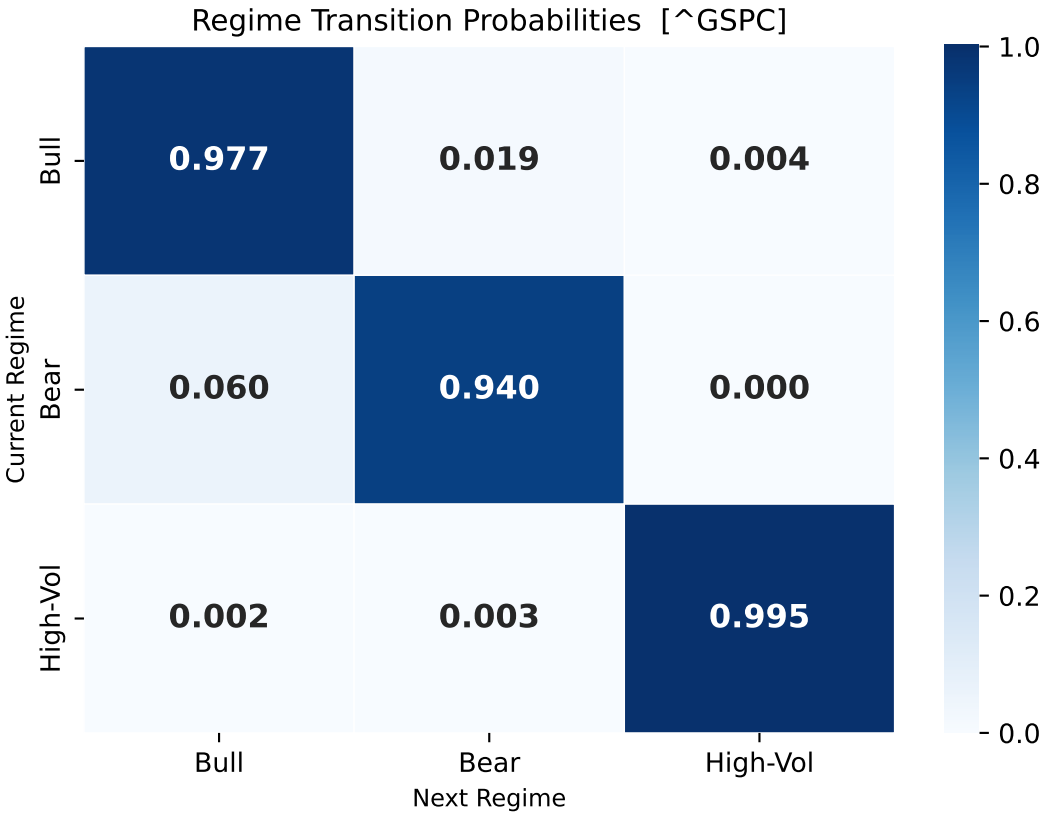


HMM with 3 hidden states fit on [log-return, Δ%VIX]. States decoded via Viterbi algorithm and labelled Bull / Bear / High-Vol by ranking mean daily return.

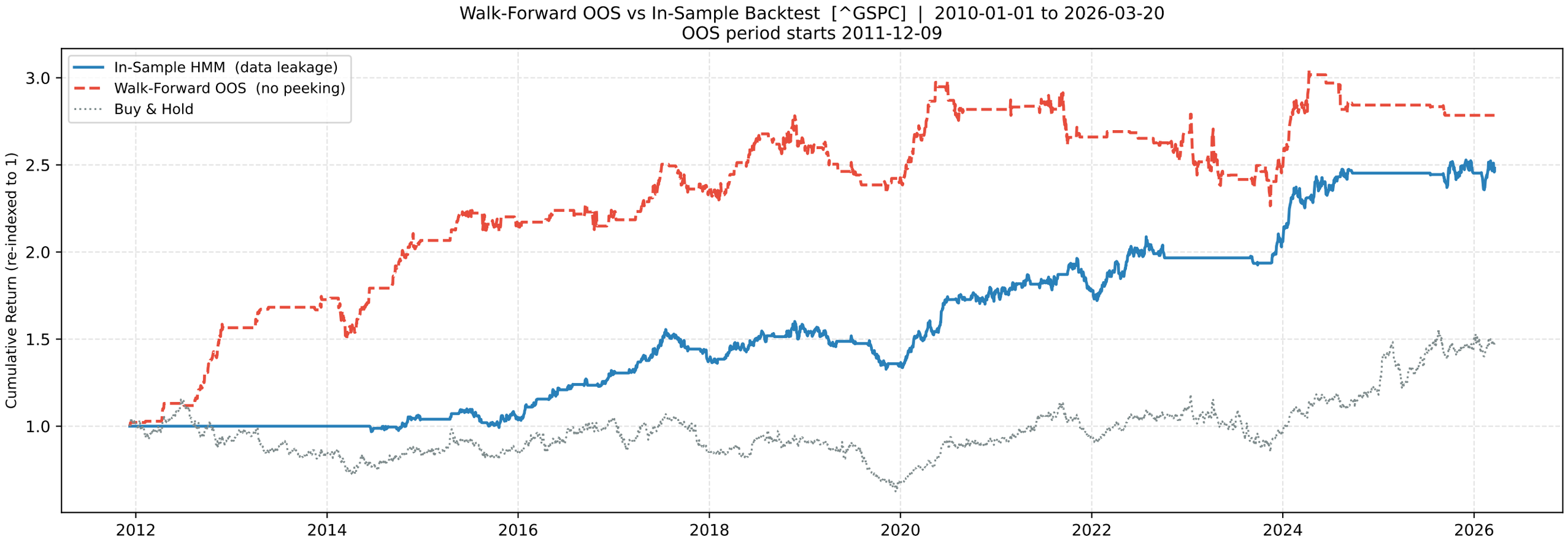




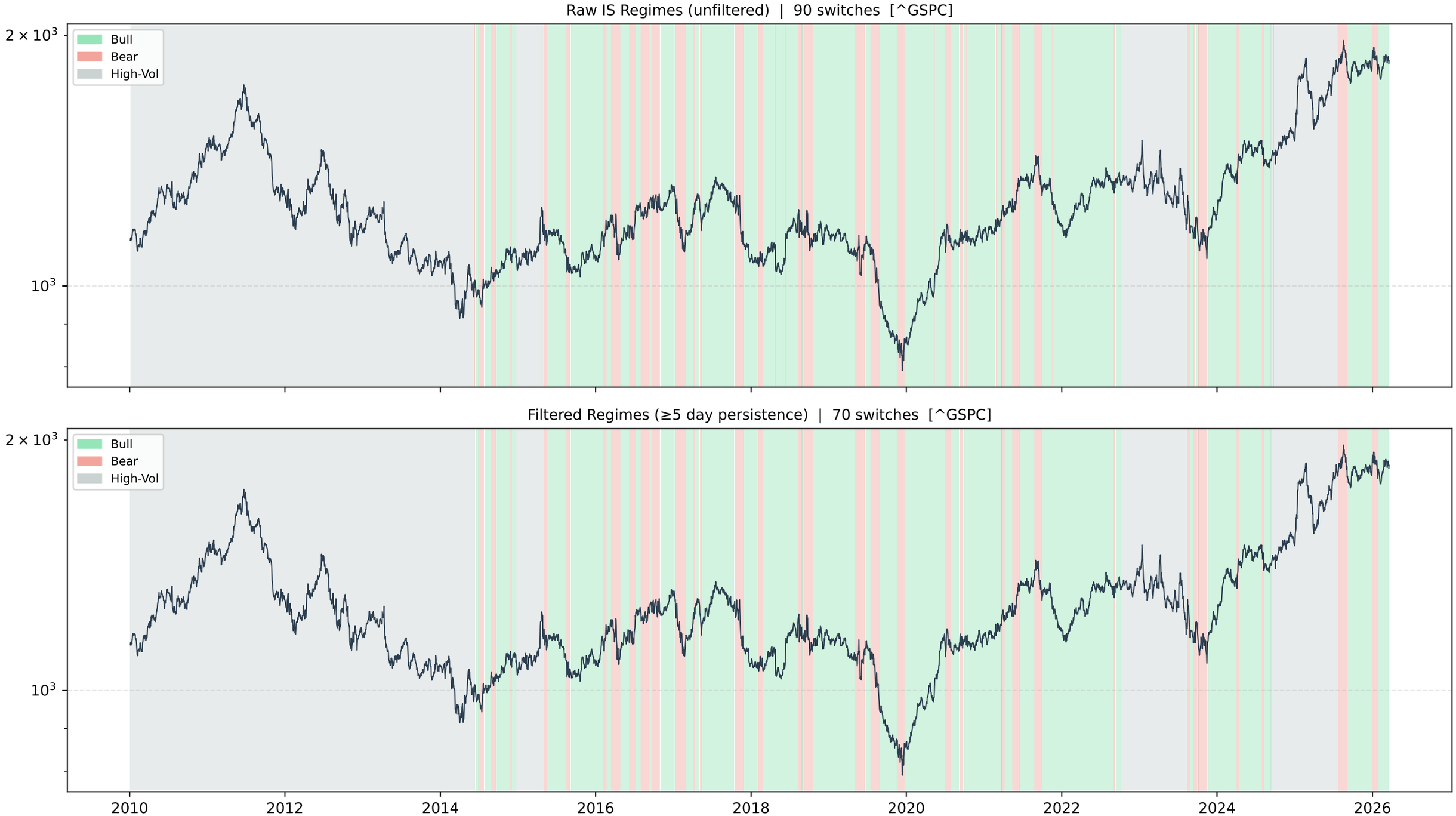
Mean daily return and realised volatility for each regime (basis points). Computed over the full in-sample training set.



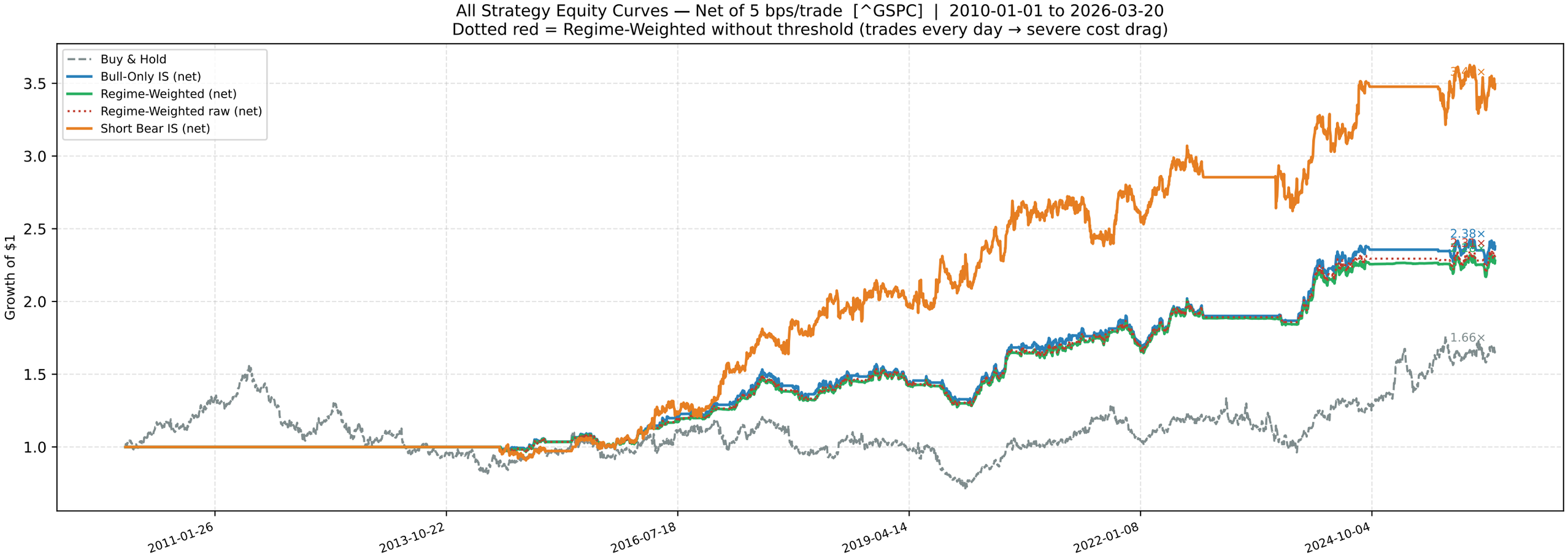
Empirical regime-to-regime transition probabilities, estimated from the Viterbi state sequence. Diagonal entries reflect regime persistence; off-diagonal entries capture typical switching paths.



Walk-forward validation: HMM re-fitted on a rolling 2-year window (504 trading days), stepping forward one month (21 days) at a time. Only the next-month predictions are retained — no future data leaks into any forecast.



Persistence filter: a regime switch is only accepted after the new state has held for at least 5 consecutive trading days. This removes high-frequency signal noise and reduces round-trips.



Four strategies compared over the full in-sample period. Bull-Only: long when IS regime = Bull, else cash.
Regime-Weighted: position = posterior P(Bull). Short Bear: +1× Bull, −1× Bear, 0 High-Vol.

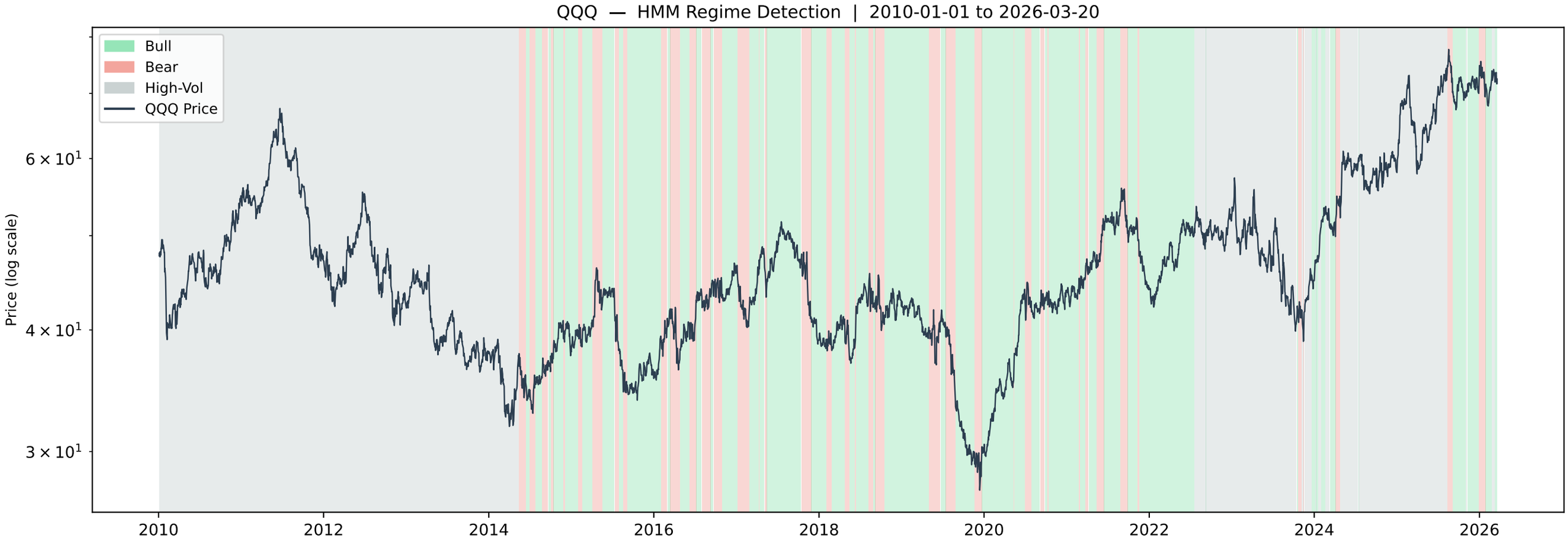
Strategy Performance Summary [^GSPC]

Data: Calibrated synthetic [^GSPC] | Regime switches: 90 raw → 70 filtered (≥5-day filter)

Strategy	Trades	Gross Return	Net Return	Gross Sharpe	Net Sharpe	Max Drawdown	Cost Drag (bps/yr)
Buy & Hold	0	3.04%	3.03%	0.18	0.18	-54.65%	0.3
Bull-Only IS	85	5.41%	5.16%	0.71	0.68	-17.33%	25.3
Bull-Only OOS	333	6.81%	5.68%	0.76	0.64	-26.36%	112.9
Persistence Filter	65	5.56%	5.37%	0.70	0.68	-17.33%	19.4
Regime-Weighted (raw)	4229	5.32%	5.00%	0.75	0.70	-17.20%	31.9
Regime-Weighted (≥5% thresh)	495	5.20%	4.91%	0.74	0.70	-16.86%	28.6
Short Bear	90	7.94%	7.45%	0.61	0.58	-14.62%	49.1

Asset: QQQ

2010-01-01 to 2026-03-20 | Calibrated synthetic [QQQ]



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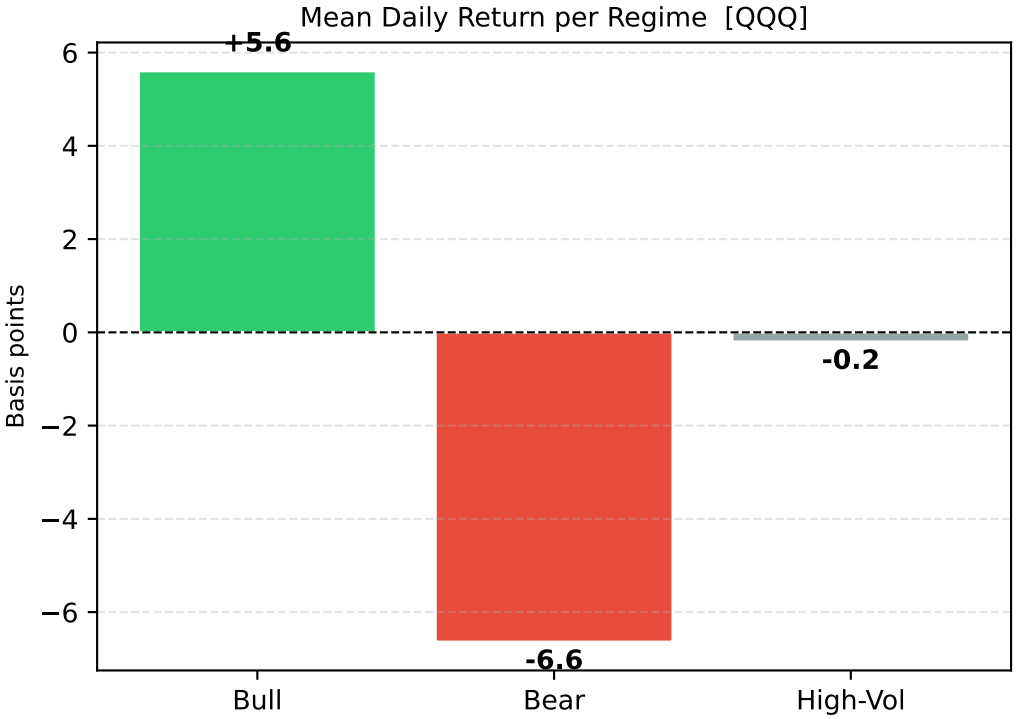

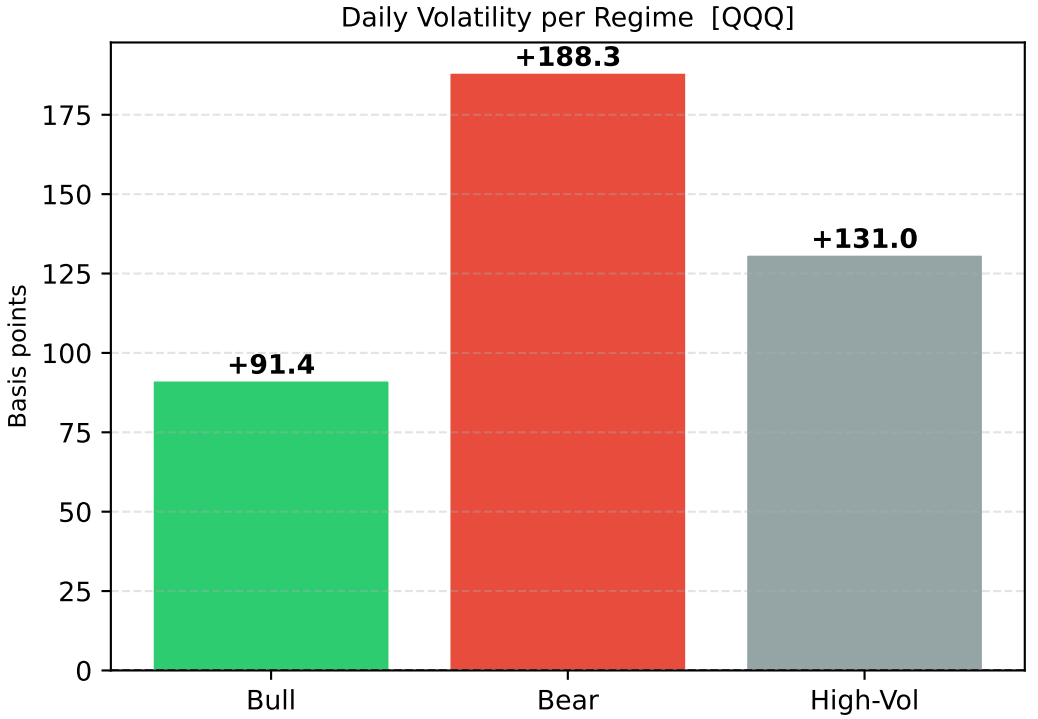

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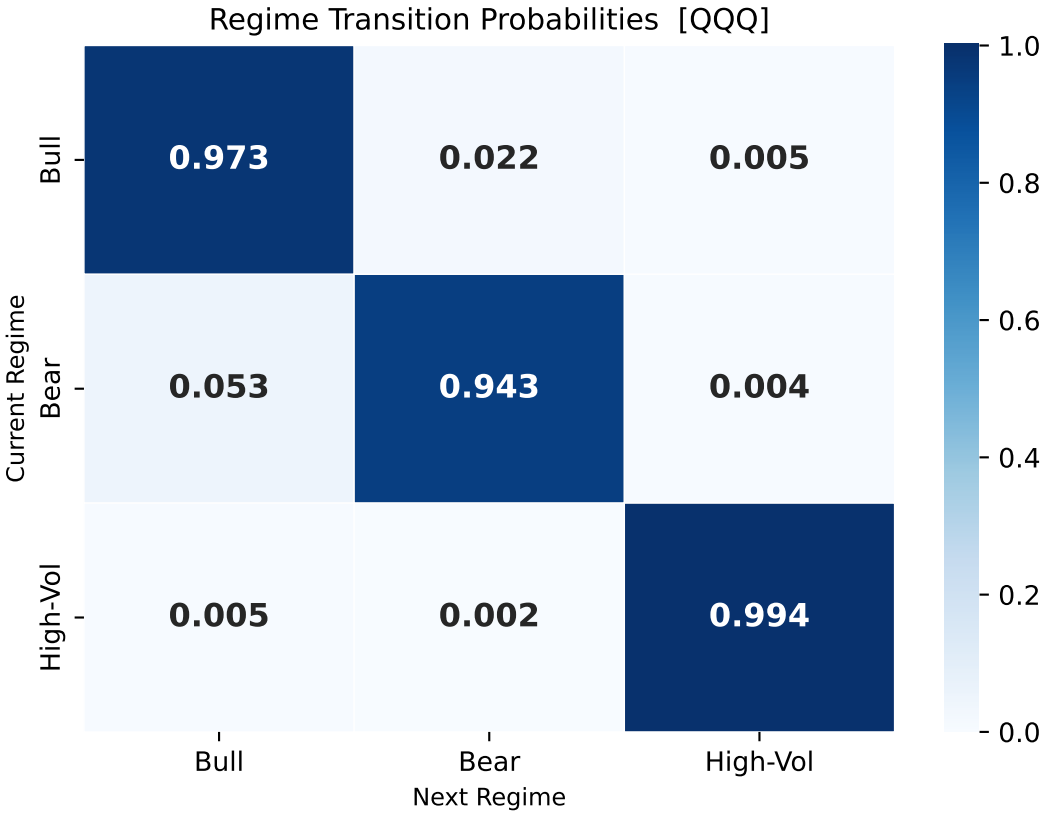

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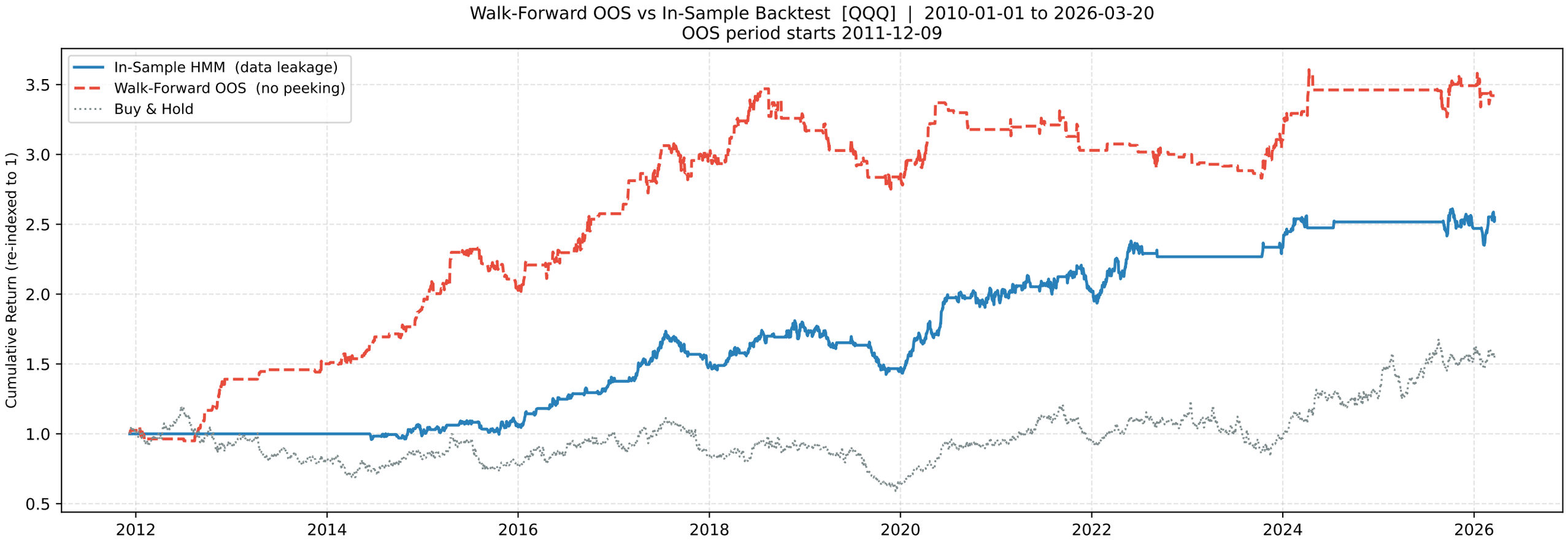
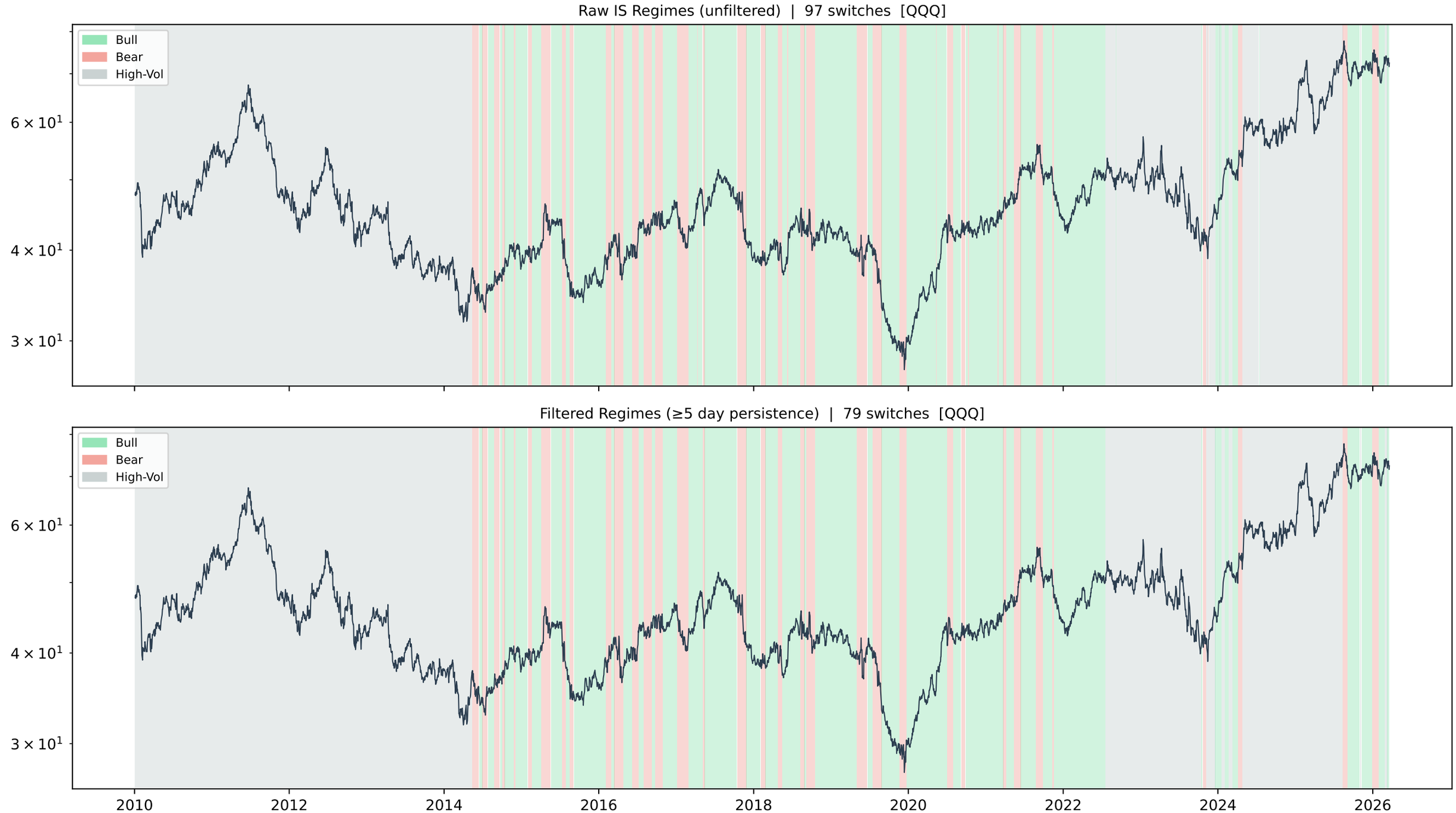
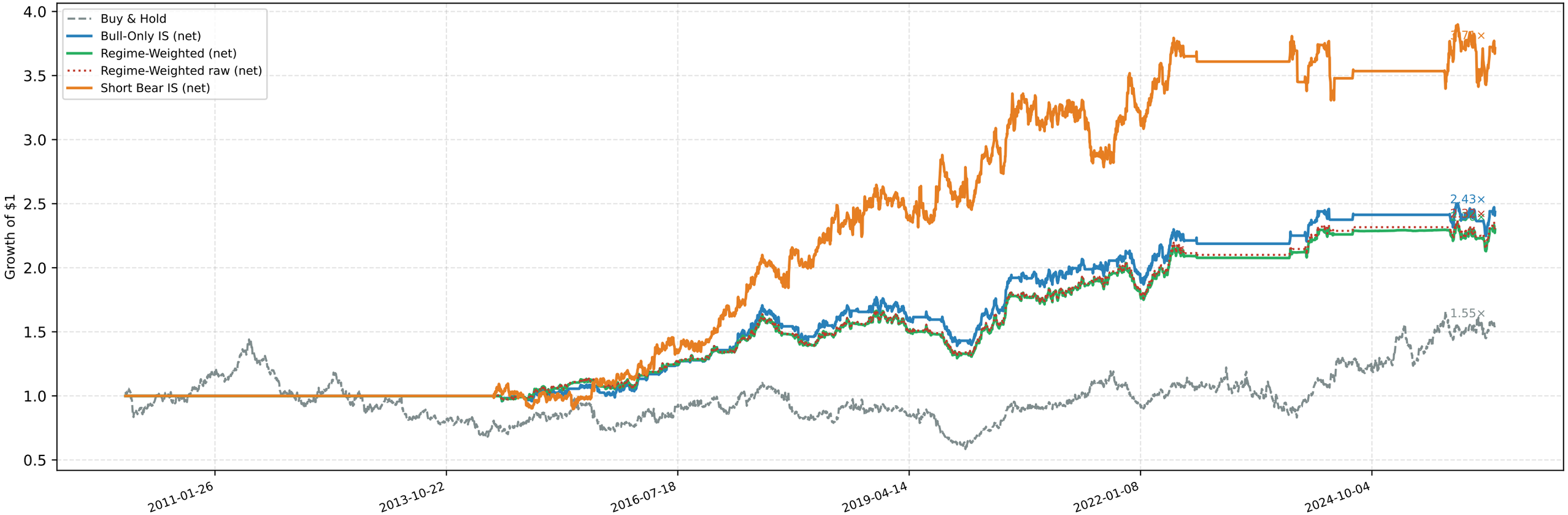


Walk-forward validation: HMM re-fitted on a rolling 2-year window (504 trading days), stepping forward one month (21 days) at a time. Only the next-month predictions are retained — no future data leaks into any forecast.



Persistence filter: a regime switch is only accepted after the new state has held for at least 5 consecutive trading days. This removes high-frequency signal noise and reduces round-trips.

All Strategy Equity Curves — Net of 5 bps/trade [QQQ] | 2010-01-01 to 2026-03-20
Dotted red = Regime-Weighted without threshold (trades every day → severe cost drag)



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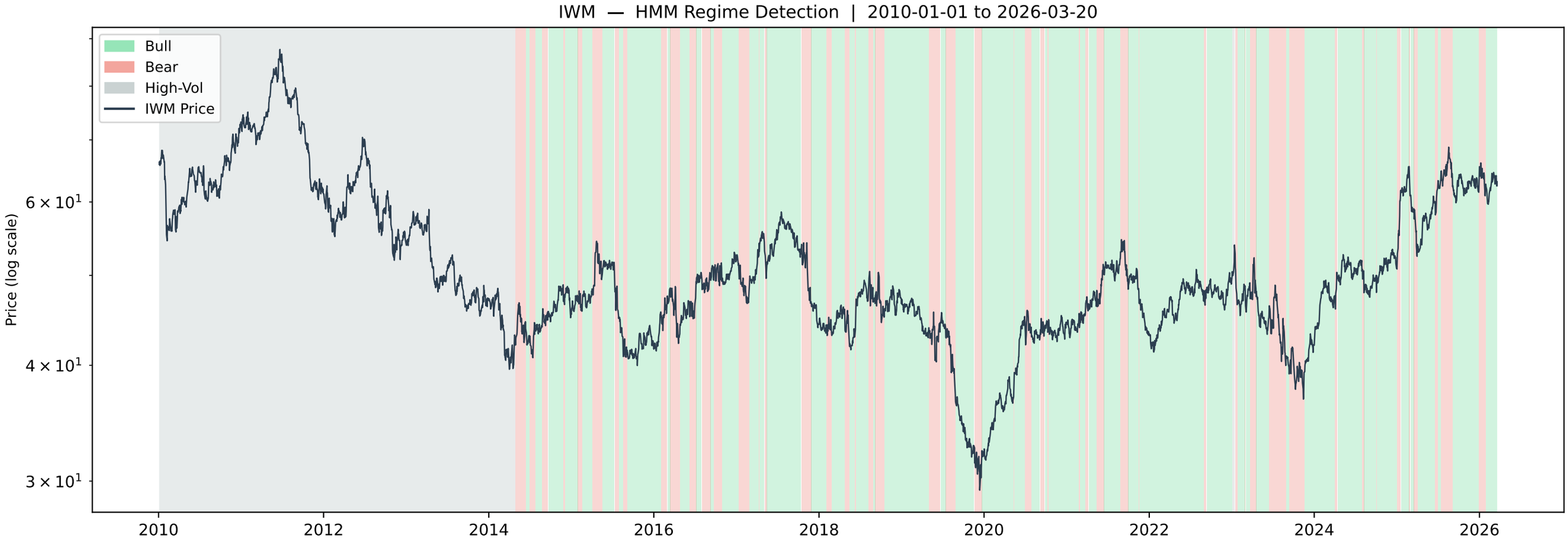
Strategy Performance Summary [QQQ]

Data: Calibrated synthetic [QQQ] | Regime switches: 97 raw → 79 filtered (≥5-day filter)

Strategy	Trades	Gross Return	Net Return	Gross Sharpe	Net Sharpe	Max Drawdown	Cost Drag (bps/yr)
Buy & Hold	0	2.59%	2.59%	0.13	0.13	-59.46%	0.3
Bull-Only IS	91	5.57%	5.30%	0.61	0.58	-21.47%	27.1
Bull-Only OOS	325	8.18%	7.08%	0.85	0.74	-22.13%	110.2
Persistence Filter	75	6.66%	6.44%	0.71	0.69	-21.47%	22.3
Regime-Weighted (raw)	4229	5.38%	5.02%	0.65	0.60	-21.98%	35.7
Regime-Weighted (≥5% thresh)	618	5.28%	4.96%	0.64	0.60	-22.27%	32.1
Short Bear	97	8.33%	7.82%	0.55	0.52	-18.18%	50.9

Asset: IWM

2010-01-01 to 2026-03-20 | Calibrated synthetic [IWM]



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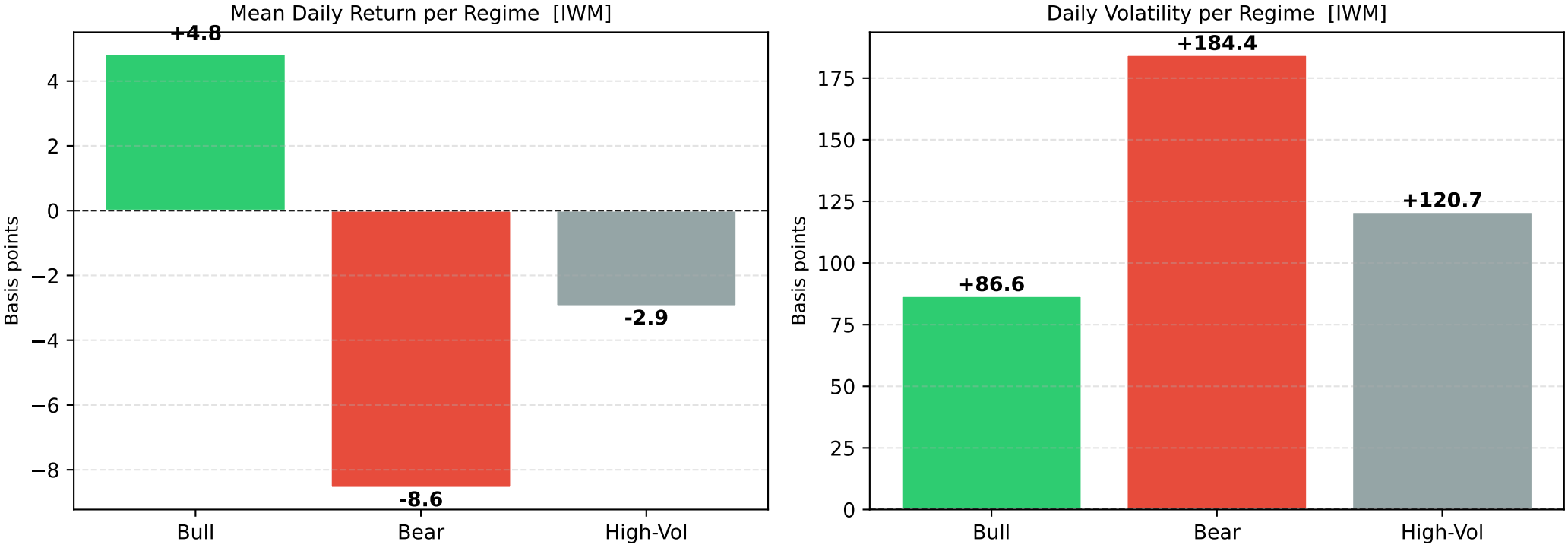

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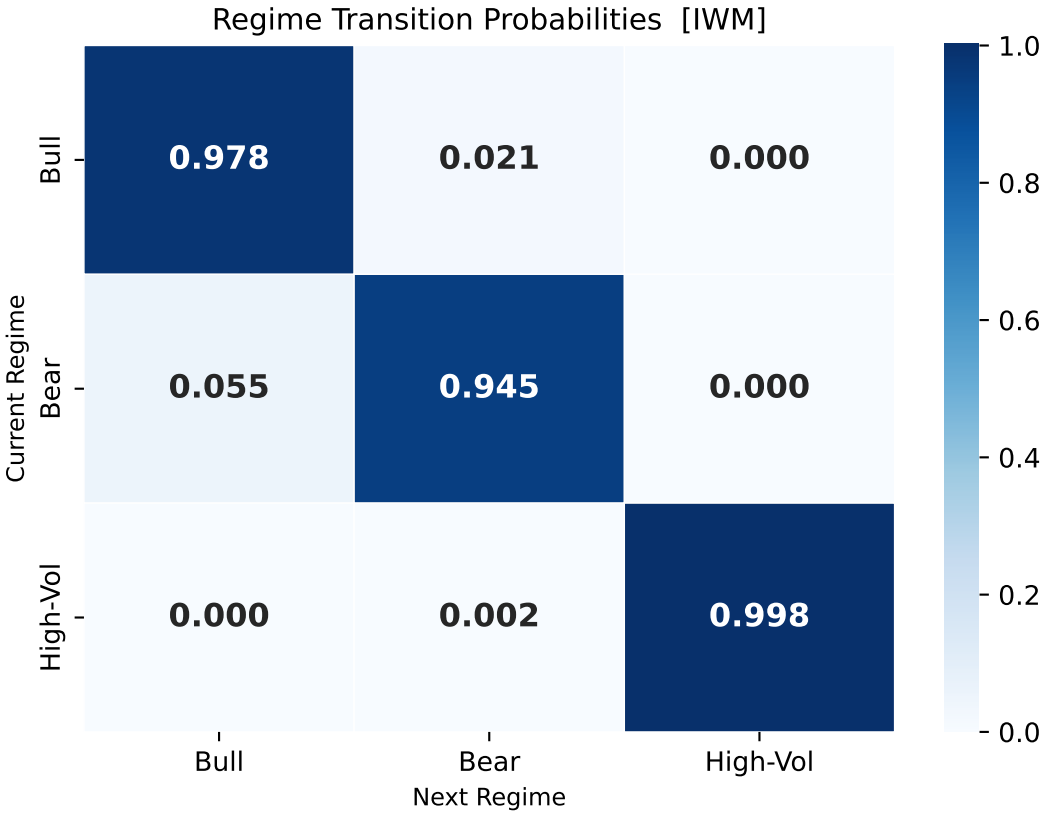

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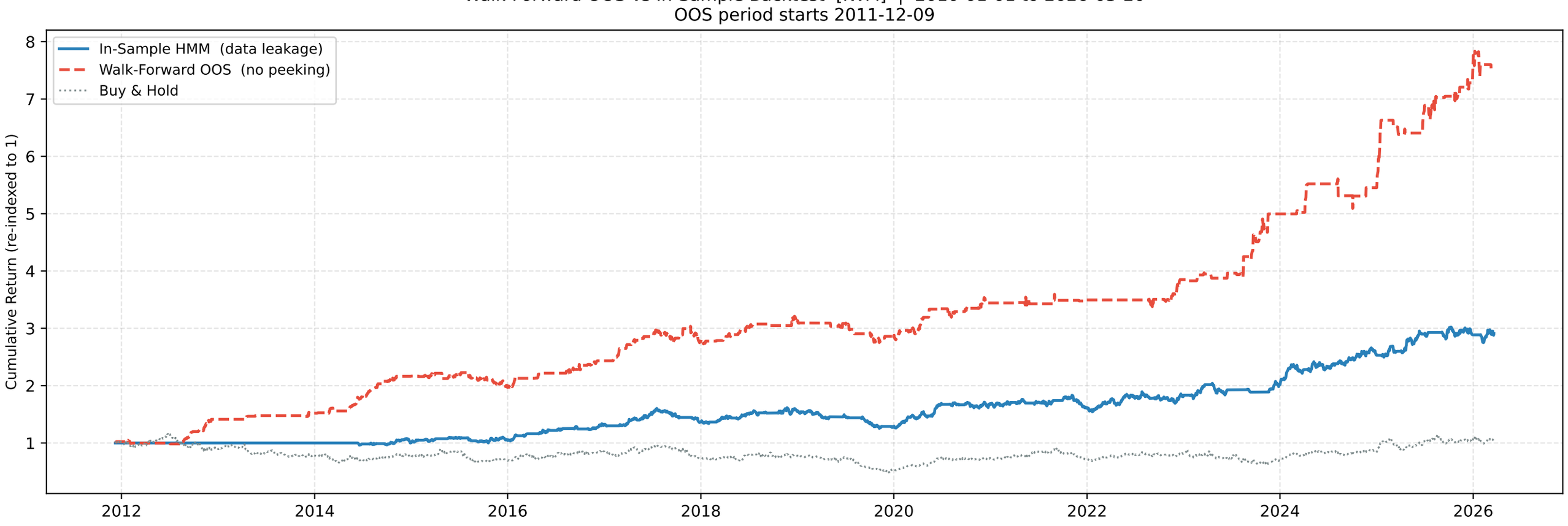

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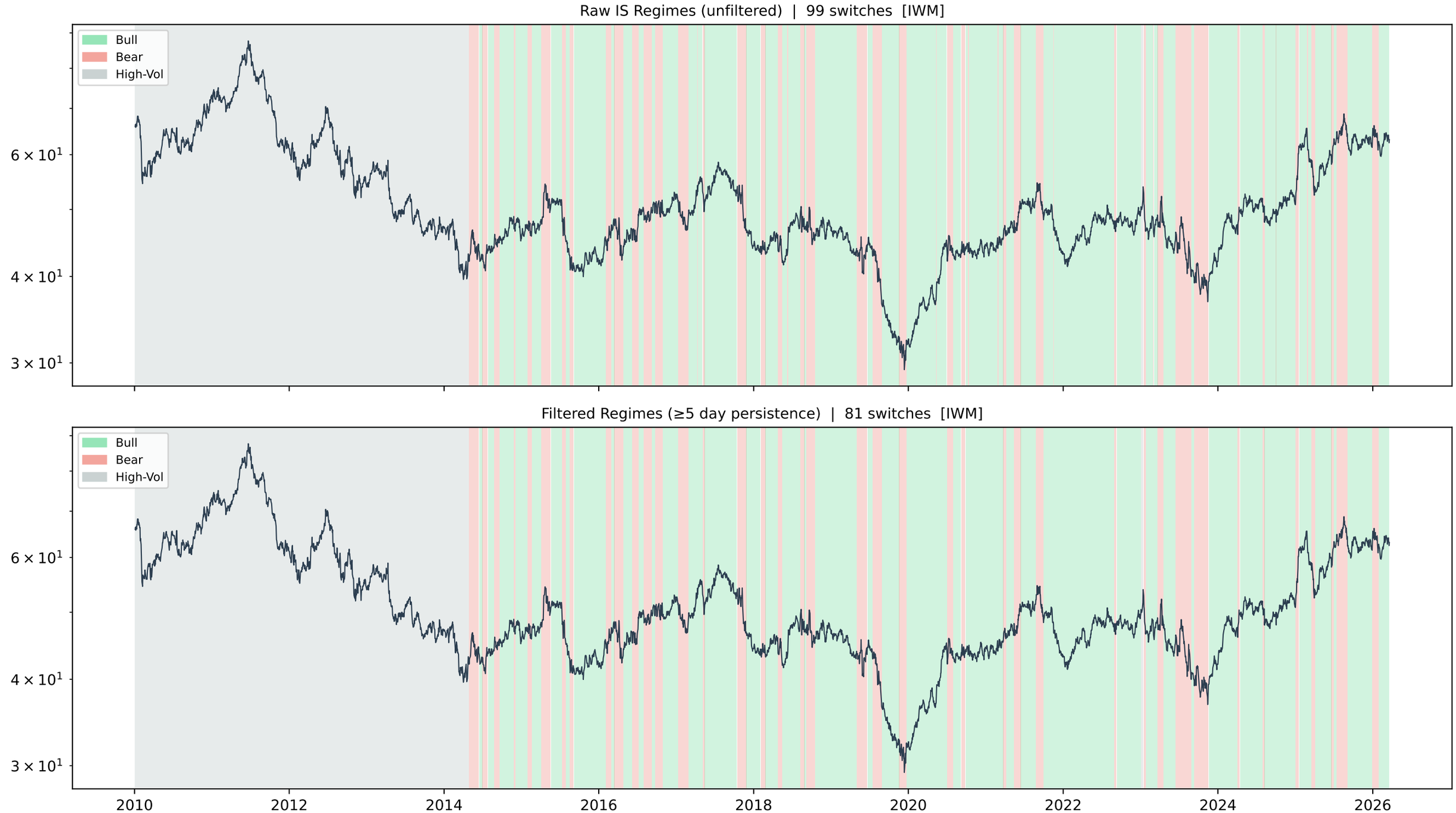

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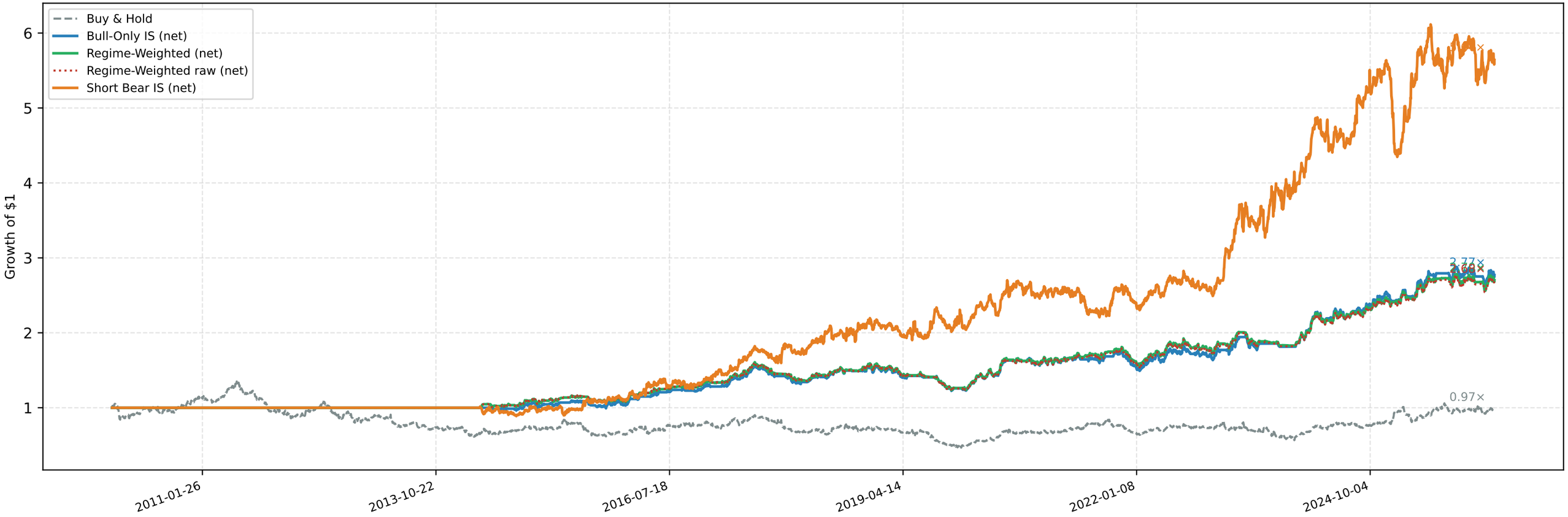

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Strategy Performance Summary [IWM]

Data: Calibrated synthetic [IWM] | Regime switches: 99 raw → 81 filtered (≥5-day filter)

Strategy	Trades	Gross Return	Net Return	Gross Sharpe	Net Sharpe	Max Drawdown	Cost Drag (bps/yr)
Buy & Hold	0	-0.16%	-0.16%	-0.01	-0.01	-66.53%	0.3
Bull-Only IS	97	6.36%	6.07%	0.64	0.61	-22.58%	28.9
Bull-Only OOS	357	13.50%	12.29%	1.41	1.29	-15.01%	121.1
Persistence Filter	79	6.48%	6.24%	0.62	0.60	-22.58%	23.5
Regime-Weighted (raw)	4229	6.28%	5.88%	0.68	0.64	-23.56%	40.2
Regime-Weighted (≥5% thresh)	725	6.28%	5.92%	0.69	0.65	-23.23%	35.8
Short Bear	99	10.89%	10.31%	0.65	0.62	-22.89%	58.1

Combined Cross-Asset Performance Summary

All strategies | 2010-01-01 → March 20, 2026

Asset	Strategy	Trades	Gross Return	Net Return	Gross Sharpe	Net Sharpe	Max Drawdown	Cost Drag (bps/yr)
^GSPC	Buy & Hold	0	3.04%	3.03%	0.18	0.18	-54.65%	0.3
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^GSPC	Short Bear	90	7.94%	7.45%	0.61	0.58	-14.62%	49.1
QQQ	Buy & Hold	0	2.59%	2.59%	0.13	0.13	-59.46%	0.3
QQQ	Bull-Only IS	91	5.57%	5.30%	0.61	0.58	-21.47%	27.1
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QQQ	Short Bear	97	8.33%	7.82%	0.55	0.52	-18.18%	50.9
IWM	Buy & Hold	0	-0.16%	-0.16%	-0.01	-0.01	-66.53%	0.3
IWM	Bull-Only IS	97	6.36%	6.07%	0.64	0.61	-22.58%	28.9
IWM	Bull-Only OOS	357	13.50%	12.29%	1.41	1.29	-15.01%	121.1
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IWM	Short Bear	99	10.89%	10.31%	0.65	0.62	-22.89%	58.1